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npj Quantum Information

https://doi.org/10.1038/s41534-026-01191-5

Article in Press

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Received: 15 November 2024

Accepted: 27 January 2026

Cite this article as: Zhuang, X.-N., Chen, Z.-Y., Xue, C. *et al.* Statistics-informed parameterized quantum circuit: towards practical quantum state preparation and learning via maximum entropy principle. *npj Quantum Inf* (2026). https://doi.org/10.1038/s41534-026-01191-5

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Statistics-Informed Parameterized Quantum Circuit: Towards Practical Quantum State Preparation and Learning via Maximum Entropy Principle

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Quantum computing offers significant potential for tackling complex problems, yet preparing quantum states from real-world data remains a critical challenge. We introduce the statistics-informed parameterized quantum circuit (SI-PQC), an approach specifically designed to efficiently prepare arbitrary statistical distributions. By leveraging statistical symmetries in data through the maximum entropy principle, SI-PQC encodes prior information with a fixed-structure circuit and tunable parameters, eliminating extensive pre-processing. This method achieves exponential resource savings in preparing mixture models, crucial for applications in statistics and machine learning. SI-PQC also supports variational learning within an optimally dimensioned training space, enhancing generalization, trainability and statistical interpretability. Numerical experiments confirm that SI-PQC can effectively prepare diverse distributions and accurately learn Gaussian mixture models, aligning closely with theoretical expectations. Applications in financial derivatives pricing and online risk analysis showcase SI-PQC's practical advantages, with substantial improvements in end-to-end quantum resource efficiency and applicability to empirical data. As a versatile and resource-efficient subroutine, SI-PQC broadens the scope of quantum algorithms, especially in real-time, data-driven fields such as finance, online machine learning, and medical diagnostics.

I. INTRODUCTION

Quantum computing has shown potential advantages in solving problems in physics, chemistry, biology, finance, and machine learning [1–6]. However, the lack of efficient input methods is one main obstacle to solving practical problems. Any promised quantum advantage will be offset if real-world information cannot be efficiently prepared [7]. Hence, quantum state preparation (QSP) is of fundamental importance. However, general QSP is difficult [8–10]. The optimal circuit depth to prepare an arbitrary n-bit quantum state is $\Theta(n)$ at a cost of exponentially many ancillary qubits [8], or $\tilde{\mathcal{O}}(\frac{2^n}{m+n}+n)$ when given m ancillary qubits [9]. As these bounds apply in the worst case where no structure or prior information about the target amplitudes is available, preparing arbitrary states necessarily incurs exponential resource growth, either in time (circuit depth) or in the number of ancillary qubits.

With prior information on the target state, quantum algorithms for practical QSP tasks have been proposed to reduce the complexity [11–20]. Such information is usually described by models, e.g., statistical distributions and their probabilistic superposition termed mixtures [21]. Existing model-based methods can prepare distributions with a formula either analytically or numerically [11–18]. Therein, prior information is translated into circuit structures and gate rotation angles via a pre-computing procedure to approximate the matrix product state (MPS) [12, 13], simulate rank-1 Hamiltonian [14], expand the Walsh series [15, 16], or implement polynomial amplitudes transformation via quantum singular value transformation (QSVT) [16–18]. There are also model-free methods to learn the distribution via a parameterized quantum circuit (PQC) or a digital Hamiltonian simulation without a known formula, e.g., Quantum Generative Adversarial Networks (QGANs) [19, 20] and flow-based generative models [22]. Therein, prior information is encoded on gate rotation angles via a pre-training procedure to minimize the loss function. Nevertheless, the above pre-processing procedures encode prior information inefficiently and implicitly. Given a statistical model with M independent parameters, one needs to update at least $\mathcal{O}(\mathrm{poly}(n))$ circuit parameters with $n \gg M$. Hence, the underlying time complexity is much higher than the QSP algorithm itself, as listed in Table I.

This quantum resource overhead is critical for practical problems. In many applications—such as financial derivatives pricing—one must model time-dependent distributions because instrument prices and volatilities evolve over time [23–28]. In such settings, model parameters vary continuously, so previously computed pre-processing artifacts rapidly become obsolete, erasing any prospective quantum speed-ups. The situation is even more challenging for mixture models, a core technique for robustness in statistics and machine learning [23, 29–34]. Here, one must prepare both a latent parameter distribution and multiple visible data distributions. Because prior information cannot be effectively shared across distinct component distributions, a straightforward combination of QSP-based subroutines with the linear

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combination of unitaries (LCU) framework [35] incurs prohibitive overhead, which is consistent with the comparison summarized in Table I.

TABLE I. **Comparison of Quantum State Preparation Algorithms.** Here, n is the qubit number, and N_θ is the size of latent parameter space for statistics mixture. The time complexity is timed by N_θ for preparing the mixture by combining previous methods and LCU. $M \ll n$ is a small constant representing the degree-of-freedom of the underlying model. See more details in Sec. 1 of Supplementary Information.

Method	Pre-process	Mixture Scaling	Ansatz Dim.
General QSP [8]	$\mathcal{O}(2^n)$	$\times N_\theta$	\
General QSP [9]	$\mathcal{O}(n2^n)$	$\times N_\theta$	\
MPS [12, 13]	$\mathcal{O}(2^n)$	$\times N_\theta$	\
Rank-1 sim. [14]	$\mathcal{O}(\frac{1}{\epsilon^2})$	$\times N_\theta$	\
Walsh [15]	$\mathcal{O}(1/\epsilon^2)$	$\times N_\theta$	\
QSVT [16–18]	$\mathcal{O}(\mathrm{poly}(n, \frac{1}{\epsilon}))$	$\times N_\theta$	\
QGAN [19]	Unclear	$\times N_\theta$	$\mathcal{O}(n)$
SI-PQC	$M \sim \mathcal{O}(1)$	$+\mathcal{O}(\log N_\theta)$	M

We argue that such pre-training and pre-computing procedures are all unnecessary if we can make appropriate use of the intrinsic symmetry in data as prior knowledge. Symmetry has long occupied a central place in physics and has been widely applied to develop quantum algorithms [36–38]. For example, point-group symmetry is employed to reduce qubit number and circuit size when simulating larger molecules [36, 37], mirror symmetry and geometric uniformity are utilized to implement multiple state discrimination [38]. However, it is still an open question to efficiently quantify and utilize symmetries within data in the context of QSP problems. Inspired by Noether's theorem, we find that observables with conservative expectations are natural candidates to characterize symmetries in data. Further, they are directly related to the model's degrees of freedom and can be efficiently derived from model parameters and data via a simple computation, which will be shown in the following sections. Moreover, the model is also uniquely determined by these observables.

In this paper, we introduce a universal model-driven QSP protocol, termed the "statistics-informed parameterized quantum circuit" (SI-PQC). SI-PQC leverages the connection between the QSP problem and the statistical model's intrinsic symmetry. By introducing the maximum entropy principle (MEP), SI-PQC can prepare arbitrary distributions and their mixtures, as depicted in Fig. 1(a). In SI-PQC, the state preparation circuit is a static structure, with tunable model parameters explicitly encoded in rotation angles. Therefore, the costly pre-computing or pre-training procedures are avoided. Moreover, we show that this static structure provides multiple advantages on various tasks. In preparation for a mixture of distributions, SI-PQC provides an exponential reduction of time and quantum resource consumption, mainly due to the reusable static structure for public model knowledge. In statistics learning tasks, SI-PQC achieves universal expressiveness and optimal learning space dimension to improve generalization ability, trainability, and interpretability, as shown in Fig. 1(b). The above advantages arise from the natural treatment of prior information, according to symmetry, as two parts: a fixed reusable part for prior knowledge from the model and a tunable part for remaining uncertainty from data. Further, we highlight the practical applicability of SI-PQC through its application to financial derivative pricing and online risk analysis of empirical data. We prove that SI-PQC is compatible with incremental computing on these essential practical problems. An end-to-end quantum resource analysis is performed to demonstrate a significant improvement in the first scenario, and an algorithm applying to empirical data is designed to expand the applicability to a more realistic regime.

The remainder of this paper is structured as follows. Firstly, we formally define the QSP problems for distributions and mixtures. Then, we introduce the maximum entropy principle and explain the main idea of SI-PQC. Next, we present the main theorems about SI-PQC for distributions and mixtures and briefly discuss the potential advantages. More explicitly, we discuss the SI-PQC-based training protocol for the Gaussian mixture model and show the results obtained with a quantum simulator. Additionally, applications on financial derivatives pricing and online risk analysis are presented with end-to-end resource analysis and numerical results. Finally, the conclusions and a discussion on possible applications of SI-PQC are presented.

II. RESULTS

A. Problem statement and framework

Formally, the main computational problems we consider are as follows:

Problem (Quantum state preparation of a statistical distribution). *Suppose a discrete statistical distribution or the discretization of a continuous statistical distribution whose probability density function (p.d.f.) is* $\mathbb{P}(x)$ *with* $\sum_x \mathbb{P}(x) = 1$. *We consider the quantum state preparation of this distribution to construct the oracle*

$$\mathcal{O}_{\mathrm{distr}} |0\rangle = \sum_x p(x) |x\rangle , \tag{1}$$

wherein $p(x) = \sqrt{\mathbb{P}(x)}$ *is the probability amplitude corresponding to the normalized p.d.f.*

Problem (Quantum state preparation of a mixture). *Suppose a statistical distribution mixture or its discretization whose p.d.f. is*

$$\mathbb{P}(x) = \sum_\theta \mathbb{P}_{\mathrm{data}}(x;\theta)\mathbb{P}_{\mathrm{para}}(\theta), \tag{2}$$

wherein $\mathbb{P}_{\mathrm{para}}(\theta)$ *is the latent distribution of the parameters with* $\sum_\theta \mathbb{P}_{\mathrm{para}}(\theta) = 1$, $\mathbb{P}_{\mathrm{data}}(x;\theta)$ *is the visible distribution of the data conditioned on the parameter* θ. *We consider the quantum state preparation of this distribution to construct the oracle*

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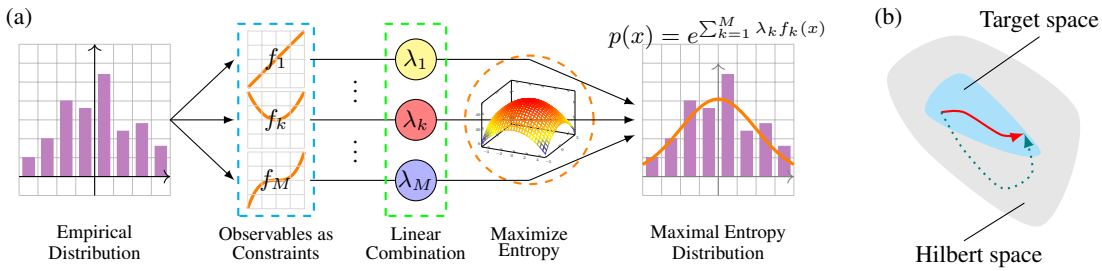

FIG. 1. **Schematic depiction of Statistics-Informed Parameterized Quantum Circuit (SI-PQC) via maximum entropy principle (MEP).** (a) MEP promises a unified formula to describe empirical data as linearly combined constraints from observables with maximized entropy. (b) SI-PQC achieves a universal expressiveness in the target space, and the output state stays therein during the optimization procedure (solid red arrow). While problem-agnostic parameterized quantum circuit output state can often deviate from the target space when varying circuit parameters (dotted green arrow).

$$\mathcal{O}_{\text{mix}} |0\rangle |0\rangle = \sum_{\theta} \sum_{x} p_{\text{para}}(\theta) p_{\text{data}}(x;\theta) |\theta\rangle |x\rangle , \quad (3)$$

wherein $p_{\text{para}}(\theta) = \sqrt{\mathbb{P}_{\text{para}}(\theta)}$ *and* $p_{\text{data}}(x;\theta) = \sqrt{\mathbb{P}_{\text{data}}(x;\theta)}$ *are the corresponding probability amplitudes.*

To address these challenges, we introduce the Statistics-Informed Parameterized Quantum Circuit (SI-PQC), a structured ansatz that directly encodes prior statistical knowledge into its design. The architecture of the SI-PQC is fundamentally derived from the Maximum Entropy Principle (MEP) [39, 40]. The MEP provides a rigorous framework for linking a distribution's underlying symmetries to its conserved statistical moments (e.g., mean, variance). This concept is analogous to Noether's theorem in physics, where symmetries in physical laws correspond to conserved quantities like energy and momentum [41]. For instance, a Gaussian distribution is the Maximum Entropy Distribution (MED) for a given mean and variance, a property reflecting its translational and scaling symmetries. Building on this foundation, recent pioneering works have demonstrated the utility of MEP for quantum state tomography and reconstruction [42–44].

We further consider the inverse problem: using SI-PQC to infer the distribution parameters from data.

Problem (Quantum-enhanced learning). *Given observed moment constraints (e.g., sample mean/variance), we consider the learning of parameters θ that minimizes the informative error and maximizes the entropy:*

$$\max_{\theta} \mathcal{S}(\rho(\theta)) \text{ subject to } \mathbb{E}_{\theta}[f_k(x)] = \mu_k, \quad (4)$$

wherein $\mathbb{E}_{\theta}[\cdot]$ is the expectation value of $f_k(x)$ given the parameterized distribution $\rho(\theta)$, f_k are the moment functions and μ_k are the empirical averages [45].

By MEP, the optimal parameters with the least informative errors can be directly evaluated from those Lagrange multipliers. MEP posits that a general distribution under mild regularity conditions can be interpreted as a MED:

$$p(x) = \frac{1}{\mathcal{N}} \exp\left\{ \sum_{k=1}^{M} \lambda_k f_k(x) \right\}. \quad (5)$$

Herein $\mathcal{N}$ is the normalization constant, M is the degrees of freedom, and λ_k are Lagrange multipliers directly associated with the optimal parameters which determine the distribution, as elaborated in Eq. (5). The generality of MED is two-fold. Firstly, any distribution with finite entropy and concave constraints is proven to be an MED [46]. In practice, M is exactly the number of independent parameters, and f_k and λ_k can be directly derived from the formula, as summarized in Supplementary Information [47]. Secondly, MED is the best unbiased guess with given observables from data when the explicit formula is not accessible [47–50]. Given only partial knowledge about the distribution law, MED minimizes the K-L divergence compared to the true distribution in the worst case, as proven in Supplementary Note 2.A.

In the SI-PQC framework, an arbitrary distribution's information is decomposed into f_k encoding the static model knowledge and λ_k representing the uncertain data information. The resulting quantum circuit can hence be separated into fixed layers for prior knowledge and parameterized layers for uncertain information. Herein, the fixed layers can be reused, compiled, and optimized. The parameterized layers can be trained as an ansatz with a minimal search space. Therefore, improved generalization ability, scalability, and trainability are realized in machine learning tasks. Note that SI-PQC only assumes a classical data-access model to implement arithmetic operations and store classical data structures. The potential quantum advantage in quantum-enhanced Monte-Carlo integration and many other quantum amplitude estimation-based algorithms will not be nullified as in a Grover-Rudolph-type algorithm, as argued in previous work [27]. As there is no strong input oracle assumption for ℓ_2-norm sampling, any further quantum advantage will not be easily dequantized by a quantum-inspired algorithm [51].

B. Maximum entropy distribution loader

In this subsection, we consider the first problem of state preparation of a distribution. Note that Eq. (5) is an exponentiation of linearly combined constraints, enabling us to construct the first type of SI-PQC, named *maximum entropy dis-*

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tribution loader (MEDL), to load a single distribution with specified parameters in three steps, as shown in Fig. 2(a) by blue, green, and orange colored blocks, respectively.

Firstly, we prepare the normalized constraint functions by operators,

$$O_{f_k} : |0\rangle \to \frac{1}{\|\vec{f_k}\|_2} \sum_{j=0}^{N-1} f_k(x_j) \, |j\rangle \, , \tag{6}$$

where $\|\cdot\|_2$ is the ℓ_2-norm of the discretized value vector $\vec{f_k} = (f_k(x_0), f_k(x_1), \cdots, f_k(x_{N-1}))$ of the underlying function at N equally spaced points x_j (the more general cases of non-equally spaced points can be transformed into this case by a composition of mapping without loss of generality). For simplicity, we will always assume equally spaced points in the following analysis. While O_{f_k} are usually non-unitary, they can be implemented on the enlarged Hilbert space by unitary dilation constructed as follows, depicted by blue blocks in Fig. 2(a)) (see detailed proof in Supplementary Note 2.B). We first prepare an efficient diagonal block-encoding of matrix $\mathrm{diag}(0, \frac{1}{2^n-1}, \frac{2}{2^n-1}, ..., 1)$ by operator

$$A = I - \sum_{j=1}^{n} \frac{2^{j-1}}{2^n} I^{\otimes(j-1)} \otimes Z \otimes I^{\otimes(n-j)} \tag{7}$$

that performs the mapping

$$|0\rangle \to \frac{1}{\mathcal{N}} \sum_{j=0}^{2^n-1} j \, |j\rangle \tag{8}$$

with $\mathcal{O}(n)$ depth and $\mathcal{O}(\log n)$ ancillary qubits. Then a non-linear transformation of amplitudes is applied with additional 2 ancillary qubits as

$$U_{f_k} : |0\rangle \, |0\rangle_{\mathrm{anc}} \to \left(\frac{1}{\|\vec{f_k}\|_2} \sum_{j=0}^{N-1} f_k(x_j) \, |j\rangle \right) |0\rangle_{\mathrm{anc}} + |\perp\rangle \, . \tag{9}$$

Note that a similar technique is also adopted in many other works, see Supplementary Note 1.B for a more detailed comparison. It is also worth noting that, to take NISQ-availability into consideration, there are also alternative methods to implement the unitary dilation by a shallow parameterized quantum circuit to approximate the block-encoding circuit [52]. SI-PQC is compatible with such a shallow parameterized quantum circuit block-encoding method and can mitigate the scalability problem due to its static structure. For the purpose of giving a systematic analysis of complexity and error, we leave this as an interesting future work. For common distributions, constraints f_k are usually elementary functions with parameters fixed to be exactly 1, as listed in Supplementary Table 2. Thus, the costly and numerical processes used in previous works can be omitted, including the unstable pre-computing of phase factors [53, 54], or training of variational quantum circuits [19].

Secondly, we inject the varying statistical information $\vec{\lambda} = (\lambda_k)_{1 \le k \le M}$ into the circuit by the linear combination of unitaries (LCU) [35]. In brief, this step creates the linear combination of constraints $f = \sum_{k=1}^{M} \lambda_k f_k(x)$ as a block-encoding of

$$U_{\mathrm{LCC}} = \sum_{k=0}^{M} \lambda_k U_k, \tag{10}$$

that performs the state preparation

$$|0\rangle \to \frac{1}{\mathcal{N}} \sum_{j=0}^{N-1} \left(\sum_{k=0}^{M} \lambda_k f_k(x_j) \right) |j\rangle \, , \tag{11}$$

wherein ancillary qubits have been omitted for simplicity. To encode the classical information separately, we build parameterized layers named sparse-state-preparation-pair, as depicted by green blocks in Fig. 2(a). The target superposition of λ_k

$$\begin{aligned} |0\rangle_M \to & \cos\frac{\theta_1}{2} |1 \underbrace{0...0}_{M-1}\rangle + \sin\frac{\theta_1}{2} \cos\frac{\theta_2}{2} |01 \underbrace{0...0}_{M-2}\rangle + \cdots \\ & + \sin\frac{\theta_1}{2} \sin\frac{\theta_2}{2} \ldots \sin\frac{\theta_{M-1}}{2} |\underbrace{00...0}_{M-1} 1\rangle, \end{aligned} \tag{12}$$

is created by the parameterized layers $P_L(\vec{\theta})$ and $P_R(\vec{\theta})$

$$\begin{aligned} & P_L(\vec{\theta}) = P_R(\vec{\theta}) \\ = & C_M X_{M-1} \circ C_{M-1} Y_M(\theta_{M-1}) \circ \cdots \\ & \circ C_3 X_2 \circ C_2 Y_3(\theta_2) \circ C_2 X_1 \circ C_1 Y_2(\theta_1) \circ X_1, \end{aligned} \tag{13}$$

wherein the subscript denotes the corresponding control and target qubit, as explicitly depicted in Fig. 3. Herein, the $\mathcal{O}(M)$ rotation angles are explicitly pre-computed to be

$$\vec{\theta} = \left(2 \arctan \sqrt{\frac{\sum_{j=k+1}^{M} 2\lambda_j \|f_j\|_\infty}{2\lambda_k \|f_k\|_\infty}} \right)_{1 \le k \le M-1} , \tag{14}$$

where $\|\cdot\|_\infty$ is the ℓ_∞-norm (See detailed proof in Supplementary Note 2.C.) The classical pre-processing procedure complexity is optimal since at least M parameters have to be updated.

Finally, we implement the imaginary time evolution of the block-encoded f via exponentiation layers (proven in Supplementary Note 2.D), depicted by orange blocks in Fig. 2(a):

$$U_{\mathrm{ME}}(\vec{\lambda}, \vec{f}) : |0\rangle \to \frac{1}{\mathcal{N}'} \sum_{j=0}^{N-1} \exp\left\{ \sum_{k=1}^{M} \lambda_k f_k(x_j) \right\} |j\rangle \, . \tag{15}$$

In particular, we apply an additional QSVT for the Taylor polynomial of degree $d = \log 1/\epsilon$ of e^x, wherein the Taylor polynomial as well as phase angles $\phi_j (1 \le j \le d)$ are obviously fixed. To summarize, we derive

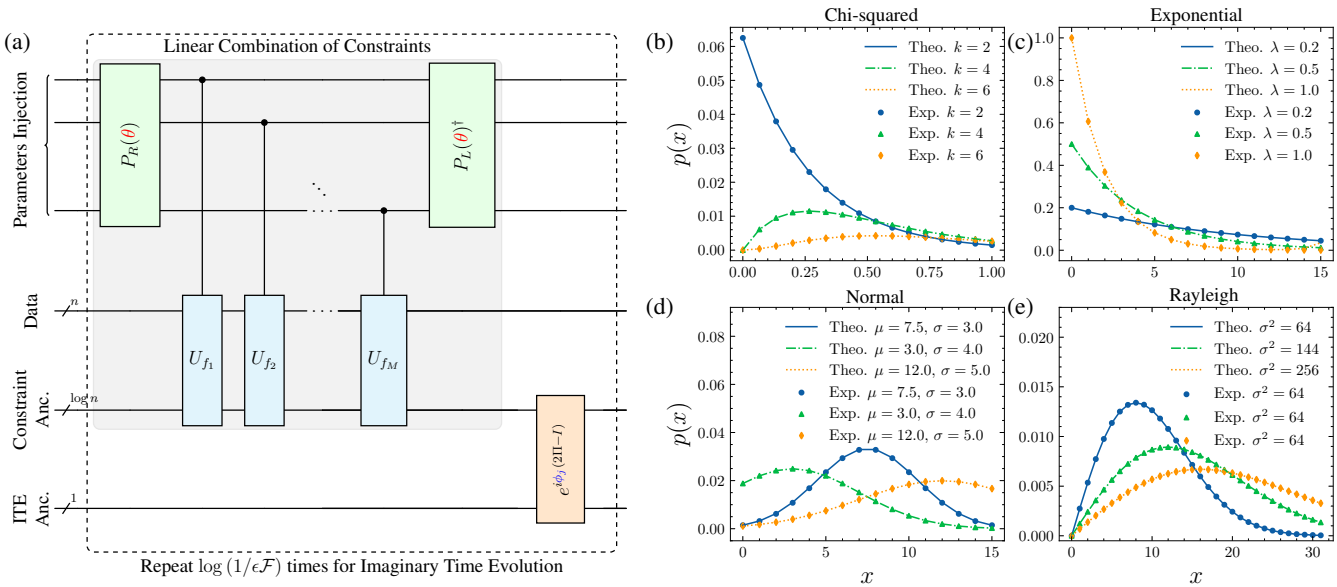

FIG. 2. **Maximum entropy distribution loader.** (a) The quantum circuit for MEDL. (b-e) The numerical results to prepare Chi-squared (upper left), exponential (upper right), normal (lower left), and Rayleigh (lower right) distributions. The scatters (Exp.) represent the experimental results, and the lines (Theo.) are the theoretical discretization of the target distributions. The approximation error ϵ (the vertical distance between scatters and lines) originates from the imaginary time evolution and converges to zero exponentially quickly as the Taylor polynomial degree $d = \log 1/\epsilon$ grows. In this experiment, we are given the perfect prior knowledge about both the distribution law and the underlying parameters.

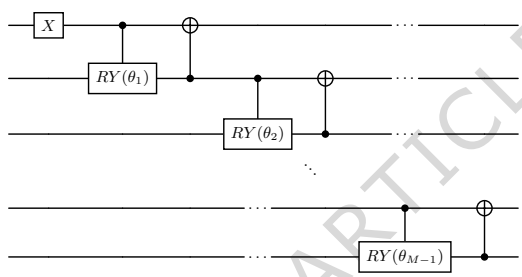

FIG. 3. **State-preparation-pair for sparse linear combination of constraints.** Each (controlled) Y-rotation gate creates a superposition of amplitudes $\sqrt{\lambda_k \|f_k\|_\infty}$ and $\sum_{j\geq k+1}\sqrt{\lambda_j\|f_j\|_\infty}$, and each CX gate resets the former qubit to state $|0\rangle$ to create a superposition of one-hot states.

Theorem (Maximum Entropy Distribution Loader). *Given an arbitrary statistical distribution with finite entropy and concave constraints defined (for discrete distribution) or discretized (for continuous distribution) on N equally spaced points $x_j (1 \leq j \leq N)$, its density probability function $p(x)$ can be viewed as a maximum entropy quantum state $|p\rangle = \sum_x \frac{1}{N}\exp\left\{\sum_{k=1}^M \lambda_k f_k(x)\right\}|x\rangle$ and can be prepared on $n = \lceil \log_2 N \rceil$ qubits approximately up to error ϵ via a statistics-informed parameterized quantum circuit with a fixed structure and trainable parameters. (1) The single-shot circuit depth without amplitude amplification is $\mathcal{O}\left(nMd_f \log\left(\frac{1}{\epsilon}\right)\right)$ with success probability to be $\mathcal{F}^2/4$, given the data space qubit number n, constraints number M, maximum constraint order d_f, filling rate $\mathcal{F} = \frac{\|p\|_{L^2}}{\|p\|_{L^\infty}}$, and ℓ_2-norm precision ϵ. The corresponding end-to-end time complexity is $\mathcal{O}\left(\frac{nMd_f \log\left(\frac{1}{\epsilon}\right)}{\mathcal{F}^2}\right)$ with $\mathcal{O}(\log n)$ ancillary qubits. (2) If amplitude amplification is employed, the circuit depth and end-to-end time complexity are both $\mathcal{O}\left(\frac{nMd_f \log\left(\frac{1}{\epsilon}\right)}{\mathcal{F}}\right)$. For distribution, $\|\cdot\|_{L^2}$ and $\|\cdot\|_{L^\infty}$ are the L^2- and L^∞-norm of the probability density function $p(x)$ in the function space, respectively.*

Herein, the circuit depth scales linearly as the qubit number n (which corresponds to the resolution and spatial discretization error of the underlying statistical model) and hence can serve as an efficient amplitude-encoded state preparation subroutine for many QML and quantum-enhanced Monte Carlo integration tasks [23–28]. The circuit depth also depends linearly on the degrees of freedom M and the maximum constraint order d_f (which characterizes the non-linear behavior of constraints), which are usually a small constant (see Supplementary Note 2 for more details). The logarithmic dependency on the precision $1/\epsilon$ comes from the imperfect rotation angles. The success probability is the cost of implementing a non-linear operation by a unitary in QSVT, and it converges to the filling rate, an intrinsic value of the distribution, as N grows. The success probability dependency on the filling rate, which is to characterize the amplitude-encoded preparation complexity of a continuous function, is equal to the optimal analytic result as far as known [14, 18]. The typical success probability for common distributions can

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be found in Table II. There are two ways to significantly reduce this overhead: Firstly, one can implement an amplitude amplification of the original circuit to reduce the end-to-end time complexity by a factor $2/\mathcal{F}$ by repeatedly using a controlled version of MEDL and a reflection operator. Secondly, one can implement a piecewise controlled-MEDL to further boost the success probability at a cost of extra control qubits in circuit compiling. The first method has been applied in this work (see the second and third columns of Table II and Supplementary Note 6 for more details). The second method is more complicated and left as future work.

We implement the Maximum Entropy Distribution Loader (MEDL) as an explicit quantum circuit and validate its output states for four distribution families: exponential, Chi-squared, normal (Gaussian), and Rayleigh (Fig. 2(b–e)). These distributions are chosen for their broad relevance to financial time series (exponential), statistical hypothesis testing (Chi-squared), machine learning (normal), and physical systems (Rayleigh). For each family, we test multiple parameter sets under the assumption of perfect prior knowledge—i.e., exact analytical forms (distribution laws) and their associated parameters (e.g., mean/variance for Gaussians, scale parameters for Rayleigh) are known. The error ϵ (depicted by the vertical distance between scatters and lines) originates from the polynomial approximation in the step of imaginary time evolution and exponentially converges to zero as the degree d grows. The experimental results exhibit close agreement with the theoretical discretization of the target distribution (Fig. 2), demonstrating MEDL's versatility in preparing distributions with conserved quantities tied to their underlying symmetries.

A key feature of MEDL is that the varying model parameters $\vec{\lambda}$ are only involved in parameterized layers, while the constraint and exponentiation layers remain invariant. Consequently, the circuit structure is static, enabling advanced circuit compilation, deep optimization, reuse, and high-precision computation of the phase factor, which is considered a primary error source in QSVT [18, 54]. Besides, note that Eq. (14) is actually a one-to-one mapping between the classical parameters $\vec{\lambda}$ and the rotation angles $\vec{\theta}$. The corresponding parameter space dimension is optimal since at least M free parameters are needed. This promises a universal expressiveness on the target problem as well as a minimum search space, enabling an efficient learning scheme to calibrate drifting distributions.

C. Mixture of distributions

In this subsection, we consider the second problem of state preparation of a mixture. Instead of a single distribution, mixtures of distributions are widely used to also consider model uncertainty, in machine learning, finance, astronomy, and robust statistics [23, 31, 33, 34, 55]. In general, a mixture can be viewed as a superposition of various distributions on visible data space, wherein the corresponding probabilities are determined by a distribution on the latent parameter space. In this subsection, we show that a mixture of distributions can be prepared via SI-PQC by reusing components in MEDL, realizing exponential resource and complexity reduction over existing quantum algorithms.

Given distributions of parameter and of data space to be $\mathbb{P}_{\text{para}}(\theta)$ and $\mathbb{P}_{\text{data}}(x;\theta)$, respectively, we can prepare the desired state defined in Eq. (2) by the second type of SI-PQC, named *Weighted Distribution Mixer* (WDM), to load the superposition of different distributions with additional model uncertainty, as pictured in Fig. 4(a). The procedure of WDM is as follows. Firstly, we prepare the quantum superposition of N_θ possible model configurations θ, each with M independent parameters, as

$$\begin{aligned}&(L_1 \otimes \ldots \otimes L_M)\,|0\rangle_{n_\theta}\\ &=\bigotimes_{k=1}^{M}\left(\sum_{j=1}^{2^{n_k^{(\theta)}}-1}\sqrt{\mathbb{P}_k^{(\theta)}(j)}\,|j\rangle\right)\\ &=\sum_{\theta}\sqrt{\mathbb{P}_{\text{para}}^{(\theta)}(\theta_1,...,\theta_M)}\,|\theta_1,...,\theta_M\rangle_{n_\theta}\end{aligned} \tag{16}$$

on the digital-encoded latent space of $n_\theta = \log N_\theta$ qubits with d_θ circuit depth. Herein, L_k is a subroutine acting on $n_k^{(\theta)} = \log N_k^{(\theta)}$ qubits to generate the kth independent parameter θ_k according to the latent distribution $\mathbb{P}_k^{(\theta)}$. Secondly, a digital-to-analog conversion, as introduced in Ref. [56], is applied to the analog-encoded latent space to create the superposition of different configurations

$$\sum_{\theta}\sqrt{\mathbb{P}_{\text{para}}^{(\theta)}(\theta_1,...,\theta_M)}\,|\theta\rangle_{\text{digital}}\,|\theta\rangle_{\text{analog}}\,, \tag{17}$$

where $|\theta\rangle_{\text{analog}}$ is the corresponding analog-encoded parameter for configuration θ. Finally, we replace the original state-preparation-pair in MEDL with this circuit to complete WDM (see Supplementary Note 3 for more details). This yields:

Theorem (Weighted Distribution Mixer of a Parametric Family). *An arbitrary distribution mixture of a parametric family $f(x;\theta)$ can be prepared as $\frac{1}{\mathcal{N}}\sum_\theta\sum_x\sqrt{\mathbb{P}_{\text{para}}(\theta)\mathbb{P}_{\text{data}}(x;\theta)}\,|\theta\rangle\,|x\rangle$ via a statistics-informed parameterized quantum circuit with a fixed structure and trainable parameters. The single-shot circuit depth is $\mathcal{O}((n_x M d_f + n_\theta + d_\theta)\log\frac{1}{\epsilon\mathcal{F}})$ with success probability being $\frac{\mathcal{F}^2}{4}$. If amplitude amplification is employed, the end-to-end time complexity is $\mathcal{O}(\frac{n_x M d_f + n_\theta + d_\theta}{\mathcal{F}}\log\frac{1}{\epsilon\mathcal{F}})$. This is given an M-dimension latent parameter space of n_θ qubits and depth d_θ, a visible data space of n_x qubits, M corresponding constraints with maximum order d_f, and a filling rate $\mathcal{F}$.*

We further constructed and validated the WDM quantum circuit for cases with partial prior knowledge, where the distribution family (e.g., Gaussian mixtures) is known, but the parameters (means, variances, weights) are uncertain. To model this uncertainty, we encode the parameter space into the latent space quantum registers, with each basis state rep-

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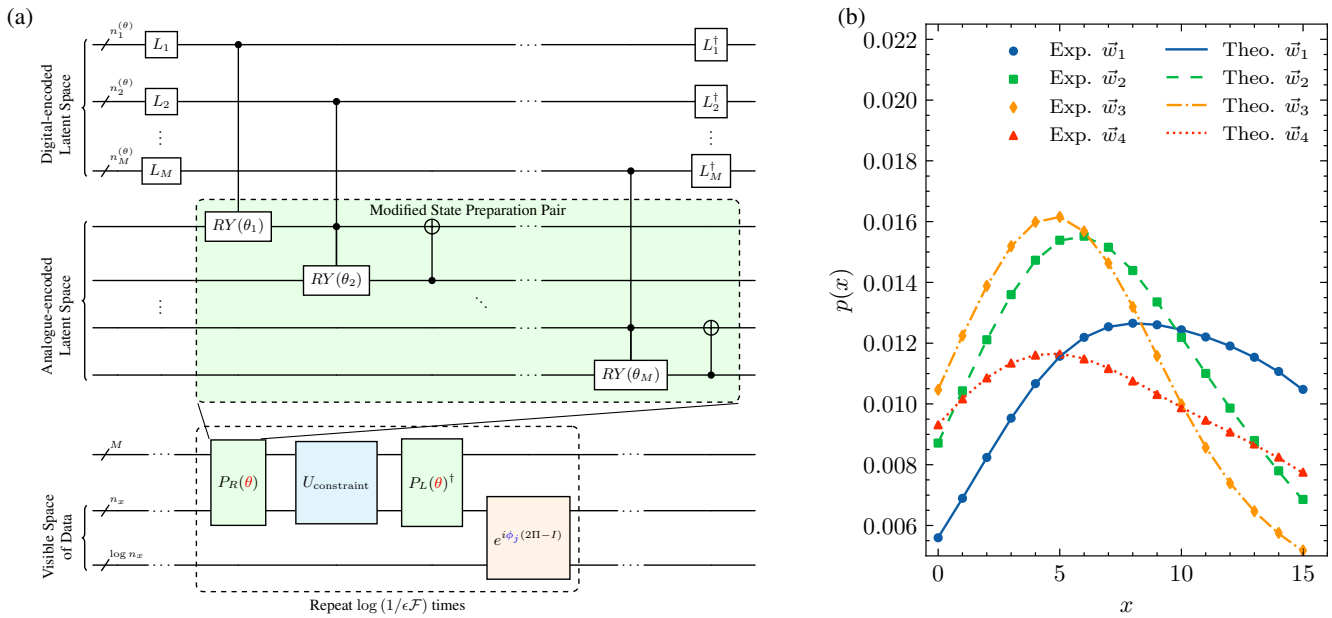

FIG. 4. **Weighted distribution mixer.** (a) The quantum circuit for the Weighted Distribution Mixer. The state preparation pair (P_L, P_R) in MEDL is substituted by an analog-encoded space controlled by the digital-encoded latent space. (b) The numerical result to prepare a 4-mixture of normal distributions with means $\mu = (1, 5, 10, 15)$, variances $\sigma^2 = (30.25, 16, 36, 36)$, and weights $\vec{w}_1 = (0.1, 0.2, 0.3, 0.4)$, $\vec{w}_2 = (0.2, 0.3, 0.4, 0.1)$, $\vec{w}_3 = (0.3, 0.4, 0.1, 0.2)$, $\vec{w}_4 = (0.4, 0.1, 0.2, 0.3)$. The scatters (Exp.) are the experimental results, and the lines (Theo.) are the theoretical discretization of the target distributions. In this experiment, we are given only partial prior information about the distribution law. The model uncertainty is described by the latent space, wherein the mixture weight corresponds to the probability of different configurations of parameters.

resenting a candidate parameter set (μ_k, σ_k^2) and its associated classical confidence weight w_k. Specifically, we demonstrate the preparation of a four-component Gaussian mixture:

$$\mathbb{P}(x) = \sum_k w_k \frac{1}{\sqrt{2\pi}\sigma_k} e^{-\frac{(x-\mu_k)^2}{2\sigma_k^2}}. \tag{18}$$

The results show a near-perfect match to the theoretical discretization of the target distribution, as depicted in Fig. 4(b). Our methods can be naturally generalized to mixtures of other distributions (see additional numerical results in Supplementary Note 7).

Note that the mixture model is not a single distribution but a superposition of a parameterized distribution family to simultaneously model the visible data space and the latent parameter space. Consequently, general QSP methods fail to directly prepare a mixture, while an LCU method requires N_θ repetitions of the whole circuit, as listed in Table I. In contrast, WDM only necessitates an additional depth of $\mathcal{O}\left((\log N_\theta + d_\theta)\log\frac{1}{\epsilon\mathcal{F}}\right)$ to implement the N_θ-mixture of parametric distributions and to model the latent distribution simultaneously. Consequently, an exponentially growing circuit depth is avoided due to the static structure and invariant output space of SI-PQC as illustrated in Fig. 5. Indeed, this exponential circuit depth reduction is a direct result of sharing the public prior information between components with different parameters. Beyond scenarios with partial parameter knowledge, the WDM framework can also be applied when the underlying distribution law is unknown. In such cases, we employ a linear combination of MEDL/WDM components, where the amplitudes can be variationally trained to learn the probability weights associated with each distribution family.

D. Quantum-enhanced statistical model learning

In this subsection, we consider the third problem of quantum-enhanced distribution learning. SI-PQC circuits, including MEDL and WDM, can create a superposition of numerous statistical models with all possible parameter configurations. This is akin to quantum brute force for statistical modeling, a concept previously discussed in the context of stochastic processes in Ref. [25]. To showcase its applicability in statistical learning, we employ a quantum-classical hybrid optimization procedure to minimize the calibration metric between the observed and predicted data $\arg\min_\theta f(Y_{\text{obs}}, Y_{\text{pre}}(x|\theta))$. The objective function is given by $\langle P_{\text{pre}}|P_{\text{obs}}\rangle = \langle 0| U^\dagger_{\text{SI-PQC}} O_{\text{obs}} |0\rangle$, which can be efficiently computed by SI-PQC. Since the hybrid training procedure is standard, we refer readers to Supplementary Note 4 for details.

We highlight the SI-PQC-based learning by Gaussian Mixture Model (GMM), a statistical model often used in machine learning and finance [23, 33, 34]. The classical GMM estimation procedure is usually costly, and a quantum method has been proposed assuming a quantum access to GMM [23].

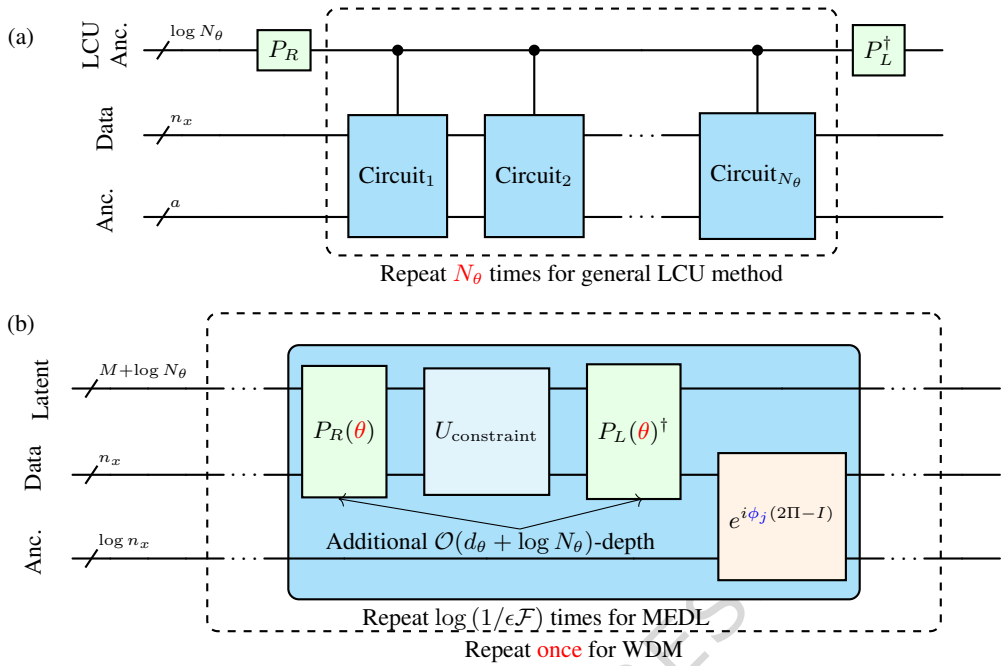

FIG. 5. **Comparison with related works for preparing a mixture of distributions.** (a) For general preparing method, an N_θ times repetition of circuits is needed for N_θ-components LCU. (b) For WDM method in this work, the circuit is implemented once with a total $\mathcal{O}(d_\theta + \log N_\theta) \log(1/\epsilon\mathcal{F})$ additional depth.

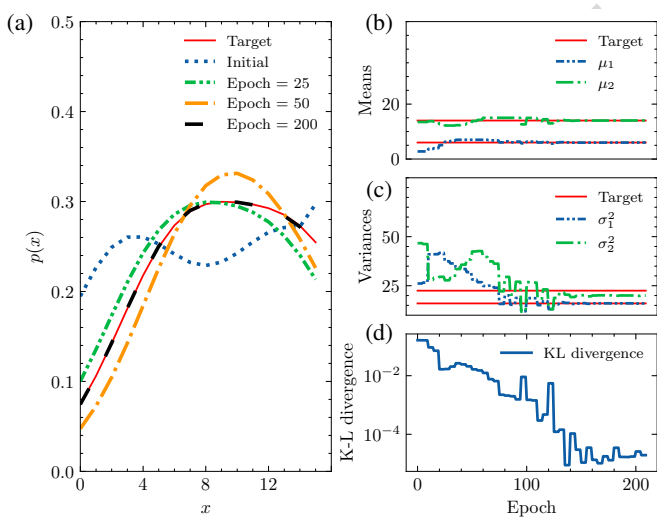

FIG. 6. **Learning Gaussian Mixture Model based on SI-PQC.** In this experiment, we are given prior knowledge about the distribution family, and its parameters and corresponding weights are learned from the data via a variational method. (a) Quantum states in the learning procedure. (b) The estimated means are extracted from the angles in the training procedure. (c) The estimated variances are extracted from the angles in the training procedure. (d) The KL divergence is used to characterize the distance from the target state.

Here, we exploit SI-PQC to train GMM without requiring quantum access.

In our numerical experiments, we assume partial prior knowledge: specifically, the distribution family is known, while its parameters and mixture weights are treated as unknown. To determine these unknown parameters, we employ a variational learning protocol compatible with MEP. This approach allows us to extract information proportional to the number of free parameters, ensuring efficient learning while maintaining physical constraints. The protocol minimizes the loss function $\mathcal{L}(\theta) = 1 - \mathcal{F}(\rho(\theta), \rho_{\text{data}})$, where $\rho(\theta)$ is the quantum state parameterized by $\theta = (\mu_k, \sigma_k^2, w_k)$, ρ_{data} is the observed (target) state, and $\mathcal{F}$ denotes the Uhlmann-Jozsa fidelity. The target mixture of two Gaussian distributions of means $\mu_1 = 6, \mu_2 = 14$, variances $\sigma_1^2 = 16, \sigma_2^2 = 22.5$, and weights $w_1 = 0.45, w_2 = 0.55$ is depicted by the red dashed line in Fig. 6(a). We started from a randomly initialized state (colored blue) and learned the final state (colored black). Notably, SI-PQC enables the simultaneous learning of classical model parameters and the quantum state. Figs. 6(b) and (c) show the estimated model parameters converging to the target means and variances, respectively. This illustrates enhanced interpretability compared to other variational-based quantum machine learning protocols. The experimental results agree very well with the target distribution, as characterized by the KL divergence and fidelity relative to the target distribution, depicted in Fig. 6(d). Note that the training protocol of SI-PQC is not limited to any specific distribution; we demonstrate the training of an exponential mixture model in Supplementary Note 7.

From a QML perspective, SI-PQC-based learning can improve generalization ability, trainability, scalability, and interpretability. Consider the generalization ability character-

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ized by the generalization error, defined as the expected error on a testing set $\mathcal{S}'$ minus the averaged error on the training set $\mathcal{S}$:

$$\mathbb{E}\left[\text{loss}(x', y')|x', y' \in \mathcal{S}'\right] - \mathbb{E}\left[\text{loss}(x, y)|x, y \in \mathcal{S}\right]. \quad (19)$$

In Ref. [57], it is proven that a QML model's generalization error is bounded by

$$\mathcal{O}\left(\sqrt{\frac{N_{\text{para}} \log N_{\text{repeat}}}{N_{\text{training}}}}\right), \quad (20)$$

where N_{para} is the number of independent parameterized gates in the QML model, N_{repeat} is the repetition time for each independent gate, and N_{training} is the training set size. Note that SI-PQC contains only $N_{\text{para}} = M \ll n$ independent gates and the repetition time $N_{\text{repeat}} = \mathcal{O}(\log \frac{1}{\epsilon\mathcal{F}})$ is independent of qubits number n. In contrast, previous work contains at least $N_{\text{para}} = \mathcal{O}(kn)$ independent parameterized gates with k layers. Therefore, SI-PQC-based learning requires significantly fewer training samples to suppress the generalization error below a given threshold. Further, we consider scalability and trainability problems for QML. For a general QML model with N_{para} independent parameterized gates, at least $\mathcal{O}(N_{\text{para}})$ expectations have to be evaluated to derive the gradient [58]. Hence, the improved scalability of SI-PQC-based learning follows directly from the need for fewer measurements. Furthermore, since the output space for SI-PQC is restricted exactly to the target Hilbert space with optimal dimension of ansatz, a mitigation of barren plateaus is rational to assume when compared to more general variational-based QML algorithms [7, 59, 60]. Specifically, since there is an explicit method to reconstruct classical model parameters from quantum circuit parameters, the learning results can also be regarded as having improved statistical interpretability. This is of particular importance in many practical scenarios such as financial derivatives pricing and gene clustering [29, 61–63].

While the above discussion focuses on statistical model learning, it can be extended to other quantum machine learning algorithms employing SI-PQC. For example, it can be directly applied to analyze the intermediate quantum states appearing in iterative quantum algorithms, such as solving partial differential equations or training quantum linear algebra-based deep neural networks [64]. Instead of costly quantum state tomography, some information can be extracted more efficiently by utilizing SI-PQC.

E. Applications

In this subsection, we highlight the efficiency of the SI-PQC-based state preparation protocol by addressing the financial derivatives pricing and Greek evaluation problem, and by performing the quantum-enhanced online risk analysis of empirical data.

Generally, financial derivatives are contracts on an underlying financial instrument with a payoff function f that depends on the underlying asset price $\mathcal{S}$. The Greeks are gradients representing the sensitivity of the derivative price with respect to changes in parameters, e.g., spot price $\mathcal{S}$ and volatility σ. Financial derivatives pricing and Greek evaluation are central yet time-intensive tasks in practical financial engineering. Fortunately, quantum-enhanced Monte Carlo integration (QMCI) has been proposed to solve this problem with a quadratic speed-up [24–28]. In the QMCI framework, an initial quantum state is prepared to encode the underlying financial instrument price distribution $\mathcal{S}$ via an n-qubit operator U_{prep}:

$$|0\rangle \to |\mathcal{S}\rangle = \sum_{x=0}^{2^n-1} \sqrt{\mathcal{S}(x)}\,|x\rangle\,, \quad (21)$$

Then quantum arithmetic or quantum signal processing subroutines U_{evaluate} then follow to compute the payoff function $f(\mathcal{S})$ as

$$|\mathcal{S}\rangle\,|0\rangle \to \sum_{x=0}^{2^n-1} \sqrt{\mathcal{S}(x)}\,|x\rangle\,(\sqrt{1-f(x)}\,|0\rangle + \sqrt{f(x)}\,|1\rangle). \quad (22)$$

Finally, the quantum amplitude estimation (QAE) is executed on an additional m-qubit register by a sequence of controlled versions of the above operators to derive the expected price $\mathbb{E}[f(s)]$ with a quadratic speed-up as desired [65].

Considering canonical quantum computational financial derivatives pricing algorithms [24–28], we employ the ancillary qubit number and the time-to-solution (TTS) as metrics, defined as

$$TTS = t_{\text{pre-process}} + (T_{\text{prepare}} + T_{\text{evaluate}}) \times N_{\text{repeat}} t_Q, \quad (23)$$

to evaluate the end-to-end resource requirement. Therein, $t_{\text{pre-process}}$ is the time complexity for classical pre-processing, T_{prepare} and T_{evaluate} are the T-depth for state preparation and payoff function evaluation, respectively, N_{repeat} is the query complexity for state preparation and payoff evaluation within QMCI framework, and t_Q is the fault-tolerant clock time of the quantum processor. Notably, the first three factors all depend on the financial instrument price sampling qubit number n, while the query complexity depends on the QAE qubit number m as $N_{\text{repeat}} \sim 2^m$.

The benchmarking TTS for QMCI with previous state preparation protocols is evaluated as follows:

$$TTS_{\text{benchmark}} = \text{Poly}_1(n) + 2^m t_Q(\text{Poly}_2(n) + T_{\text{evaluate}}), \quad (24)$$

wherein $\mathcal{O}(n)$ ancillary qubits are employed. The pre-processing time complexity and the state preparation T-count are lower bounded by $\text{Poly}_1 \in \Omega(n^2)$ and $\text{Poly}_2 \in \mathcal{O}(n^2)$, respectively. Herein, the pre-processing time complexity originates from training a PQC to generate a standard Gaussian distribution, and the ancillary qubit number's linear dependence on n and the state preparation's quadratic dependence on n both stem from the costly quantum arithmetic operations used to inject varying model parameters μ, σ. In contrast, QMCI with SI-PQC state preparation enables a quan-

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tum extension of the classical online computing technique termed *incremental computing*. Model parameters, estimated from online data via classical incremental computing, can be computed with $\mathcal{O}(1)$ complexity, and then circuit parameters can also be updated within $\mathcal{O}(1)$ time. The ancillary qubit number is $\log n + 6$, and the TTS for QMCI with SI-PQC is

$$TTS_{\mathrm{SI-PQC}} = \mathcal{O}(1) + 2^m t_Q(\mathrm{Poly}_3(n) + T_{\mathrm{evaluate}}), \quad (25)$$

where $\mathrm{Poly}_3 \in \mathcal{O}(n \log n)$ is proven to have a lower T-depth. Consequently, we significantly improve TTS by utilizing SI-PQC (see Supplementary Note 5). Furthermore, consider the evaluation of the Greeks. The above pricing algorithms must be run many times with different model parameter configurations. When applying previous state preparation methods, the costly pre-processing procedure must be run many times. Since SI-PQC is a parameterized quantum circuit, we can directly exploit the parameter shift rule to derive the partial derivatives.

Risk analysis is another fundamental task in finance for quantifying potential losses. Value at Risk (VaR) is one of the most commonly used measures to estimate the maximum possible loss at a specified confidence level [66, 67]. Formally, let the asset price be a stochastic variable X with a cumulative distribution function $F(x) = \int_{-\infty}^{x} \mathbb{P}(t)dt$. The VaR at a confidence level $\alpha \in (0, 1)$ is then defined as:

$$\mathrm{VaR}_\alpha[X] = -\inf_x \{F(x) > \alpha\}. \quad (26)$$

Classically, evaluating VaR involves computationally intensive numerical integration over the loss distribution, often relying on empirical data [67] and binary search methods. Recently, quantum algorithms, such as QMCI, have been proposed to accelerate VaR evaluation [24, 68]. In this approach, the potential loss distribution is encoded within a quantum state $\sum_x \sqrt{\mathbb{P}(x)}\,|x\rangle$. A comparator is then implemented to perform

$$\sum_{x \le L} \sqrt{\mathbb{P}(x)}\,|x\rangle\,|0\rangle + \sum_{x > L} \sqrt{\mathbb{P}(x)}\,|x\rangle\,|1\rangle, \quad (27)$$

followed by QAE to quadratically accelerate the evaluation of $\mathbb{P}[X \le L]$. Finally, the binary search algorithm is employed to find the minimal L_α satisfying $\mathbb{P}[X \le L] \ge 1 - \alpha$ [24].

In practice, while empirical distributions are preferred, encoding arbitrary distributions directly into quantum states remains challenging. Leveraging the SI-PQC preparation protocol, we propose a framework for VaR evaluation using real-time empirical data. With live risk data, both empirical and estimated distributions are evaluated on a sliding window, as shown in Fig. 7(a). Monitored observables are updated incrementally with each new data tick. Specifically, slowly varying observables can serve as conservative constraints to approximate the empirical distribution, supported by the maximum entropy principle, as depicted in Fig. 7(b). Note that prior knowledge about symmetries can be data-driven, while the distribution law remains unknown. The amount of information inferred from data is determined by the number of monitors that successfully capture conserved quantities. The SI-PQC protocol allows prior knowledge to inform circuit parameters, translating expectations into quantum states (as illustrated in Fig. 7(c)). Finally, SI-PQC-based QMCI is applied to compute the quantum VaR. This algorithm is summarized in Alg. 1.

In the numerical experiments, data were generated randomly over 33000 timestamps from a one-dimensional Maxwell-Boltzmann distribution $\mathbb{P}(x) = \sqrt{\frac{2}{\pi}}\frac{x^2}{a^3}\exp\left\{\left(\frac{-x^2}{2a^2}\right)\right\}$ with varying parameter $\mathbf{a} = (4, 5, 6, 6, 5.5, 6, 7, 6, 5, 5.5, 6)$, where each value was applied over 3000 ticks. The first 1000 ticks of data and the estimated density function are displayed in Fig. 7(a). The monitor window length is set to 1000 to observe the expectations $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathbb{E}[\ln X]$. As shown in Fig. 7(b), $\mathbb{E}[X]$ and $\mathbb{E}[\ln X]$ vary slowly, and thus serve as candidates for conservative constraints. Based on the partial prior knowledge of conservative constraints for the maximum entropy principle, we selected an appropriate SI-PQC circuit to approximate the historical data using $n = 5$ qubits. The corresponding measurements closely replicate the empirical distribution, as demonstrated in Fig. 7(c). In Fig. 7(d), quantum-enhanced VaR estimates at different timestamps (grouped by 100 for simplicity) and for confidence levels $\alpha = 0.95, 0.85, 0.05$ are indicated by diamonds, stars, and triangles, respectively. Exact VaRs from empirical data serve as benchmarks, represented by solid blue, dotted green, and dashed orange lines. These results demonstrate that SI-PQC state preparation can efficiently and accurately approximate empirical distributions, expanding the applicability of quantum risk analysis to more realistic scenarios involving empirical data.

Algorithm 1 Quantum Online Risk Analysis of Empirical Data.

Input: Risk time series X, window length L, confidence level α
Output: Estimated Value at Risk $\mathrm{VaR}_\alpha[X]$
Initialize data queue X
Initialize monitor of observables, e.g., $\mathbb{E}[X]$, $\mathbb{E}[\ln X]$
Initialize queue pointers x_{-1}, x_1 for the latest and the farthest ticks, respectively.
while new tick data X_{new} **do**
  for observables $f(\cdot)$ **do**
    Update expectation $\mathbb{E}[f] \leftarrow \mathbb{E}[f] + \frac{f(X_{\mathrm{new}}) - f(x_1)}{L}$
  end for
  Update SI-PQC circuit parameters θ using Eq. (14)
  Generate quantum states using SI-PQC
  Apply QAE procedure to evaluate VaR
  Update pointer $x_{-1} \leftarrow X_{\mathrm{new}}$
  Update pointer x_1 to discard the farthest tick
end while
Compute estimated Value at Risk $\mathrm{VaR}_\alpha[X]$

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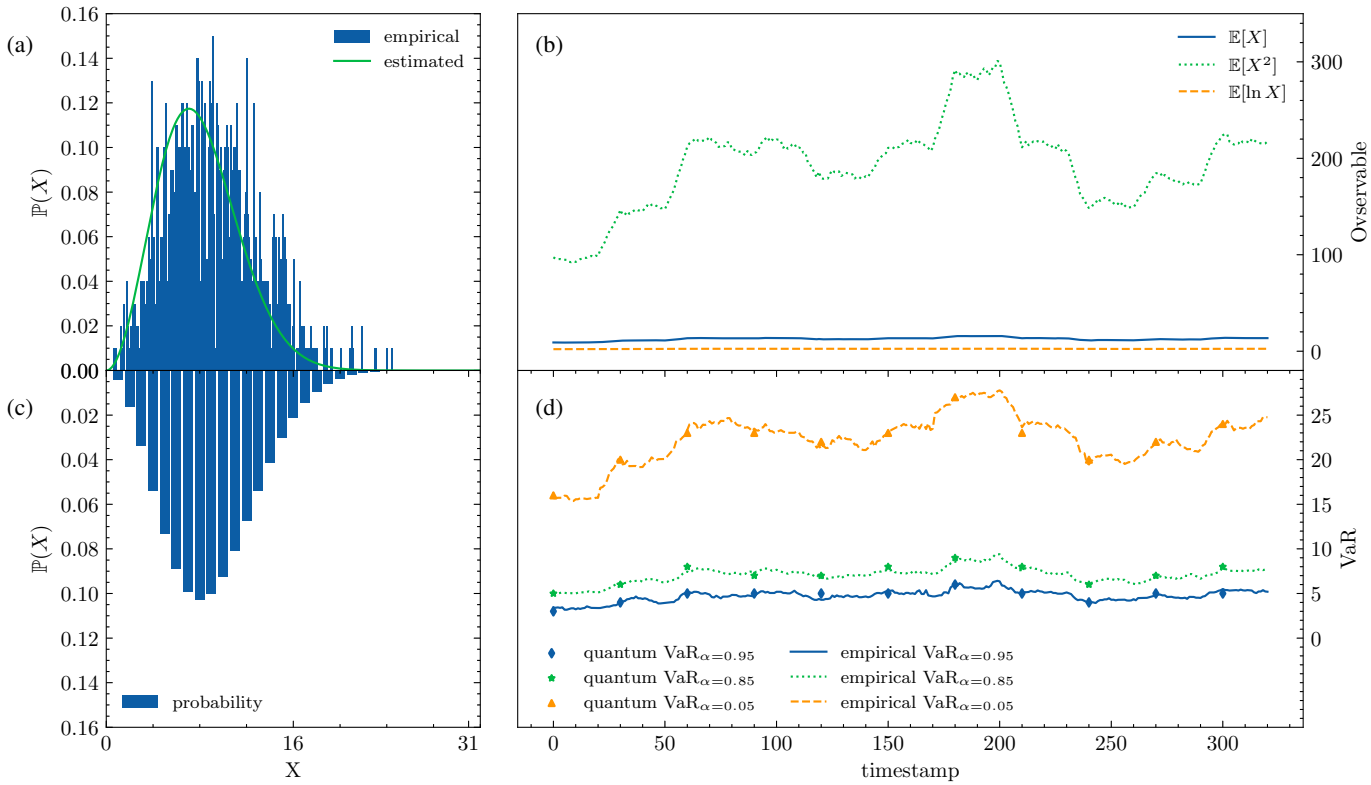

FIG. 7. **Quantum Risk Analysis based on SI-PQC.** (a) Empirical data and estimated risk distribution, X is the underlying asset price, and $\mathbb{P}(X)$ is the empirical/estimated probability. (b) Monitored observables that can be efficiently derived by classical incremental computing. Herein, the observables $\mathbb{E}[X]$ and $\mathbb{E}[\ln X]$ vary slowly and can be regarded as conservative constraints, while $\mathbb{E}[X^2]$ changes violently. Henceforth, one can model this empirical stochastic variable by a Maxwell-Boltzmann distribution $\mathbb{P}(x) = \sqrt{\frac{2}{\pi}}\frac{x^2}{a^3}\exp\left\{\left(\frac{-x^2}{2a^2}\right)\right\}$. (c) Quantum states are prepared by SI-PQC using prior information from classical monitors. The model parameters can be directly estimated as $a = \sqrt{\frac{\pi}{8}}\mathbb{E}[X]$. (d) The solid, pointed, and dashed lines are the empirical Value at Risk (VaR)s given levels $\alpha = 0.95, 0.85, 0.05$, respectively. The diamond, star, and triangle markers are the quantum VaRs given levels $\alpha = 0.95, 0.85, 0.05$, respectively.

F. End-to-end resources

We have analyzed the quantum resources for preparing representative distribution families, as summarized in Table II. Herein, end-to-end T-gate count dominates the physical qubits for the magic state factory, and the ancillary qubits mainly contribute to the logical qubits. The end-to-end T-depth determines the time-to-solution. We further analyze quantum resources required to prepare an arbitrary normal distribution and compare them with the Grover-Rudolph/Möttönen method and the re-parameterization method, as listed in Table III. The analysis suggests an optimal end-to-end quantum resource for SI-PQC.

We have also numerically tested the ancillary qubit number, T-gate count, and T-count for preparing a normal distribution, comparing these metrics with the Grover-Rudolph/Möttönen method and the re-parameterization method, as depicted in Fig. 8. The pre-trained PQC's layer number is set to $L \simeq 10$ for $n = 7$ as in Ref. [27]. We extrapolate this result assuming $L = n$, which is a common assumption. The canonical value of $\mathcal{F}$ is approximately 0.543 for $\sigma \geq 1/6$ (corresponding to the $\pm 3\sigma$ interval). For a smaller σ, the distribution can be prepared on a truncated interval to enhance the success probability. The numerical experiments suggest that the re-parameterization method and the SI-PQC method both outperform the Grover-type method, achieving an exponential reduction in the key factor n. Moreover, SI-PQC achieves an optimal spacetime overhead for large values of n.

However, it should be noted that, for small values of n that are common in practical scenarios, this slight advantage can be diminished by a larger prefactor, as depicted in Fig. 8. This can be viewed as a limitation for SI-PQC. An exciting alternative solution is to combine SI-PQC with a pre-trained PQC for block encoding of the linear function [27]. This will help reduce the quantum resources required to achieve the crossover to practical quantum advantage in the early fault-tolerant era.

Notably, besides preparing a single distribution, a promising advantage of SI-PQC is to model mixtures of parametric families. Due to the separation of fixed model structure and tunable parameters, SI-PQC can model the latent space for varying parameters with an exponential quantum resource

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TABLE II. Fault-tolerant quantum resource for representative distribution families

Family	End-to-end T-depth[1]	End-to-end T-count[2]	Ancillas
Normal	$\left((2n+8)\cdot\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+30\log_2(1/\epsilon_\theta)\right)\cdot\frac{2\ln(1/\epsilon)}{\mathcal{F}}$	$((8n+32)\lceil\log_2 n\rceil-8n+(9\lceil\log_2 n\rceil+21)\cdot\log_2(1/\epsilon_\theta)-12)\cdot\frac{2\ln(1/\epsilon)}{\mathcal{F}}$	$2\lceil\log_2 n\rceil+5$
Exponential	$\left((n+3)\cdot\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+15\log_2(1/\epsilon_\theta)\right)\cdot\frac{2\ln(1/\epsilon)}{\mathcal{F}}$	$((4n+12)\lceil\log_2 n\rceil-2n+(6\lceil\log_2 n\rceil+9)\cdot\log_2(1/\epsilon_\theta)+2)\cdot\frac{2\ln(1/\epsilon)}{\mathcal{F}}$	$2\lceil\log_2 n\rceil+4$
Rayleigh	$(2\ln\left(\frac{1}{\epsilon}\right)+1)\left[(n+4)\left(\left\lceil\log_2\frac{\log_2 n}{3}\right\rceil+2\right)+13.5\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$(2\ln\left(\frac{1}{\epsilon}\right)+1)\left[(n+5)(4\lceil\log_2 n\rceil-4)+3(\lceil\log_2 n\rceil+3.5)\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$3\lceil\log_2 n\rceil+5$
Chi	$(2\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+4)\left(\left\lceil\log_2\frac{\log_2 n}{3}\right\rceil+2\right)+7.5\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$(2\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+4)(4\lceil\log_2 n\rceil-6)+(3\lceil\log_2 n\rceil+4.5)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$3\lceil\log_2 n\rceil+5$
Chi-squared	$(\ln\left(\frac{1}{\epsilon}\right)+\frac{k}{2}-1)\left[(n+3)\left(\left\lceil\log_2\frac{\log_2 n}{3}\right\rceil+2\right)+6\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$(\ln\left(\frac{1}{\epsilon}\right)+\frac{k}{2}-1)\left[(n+3)(4\lceil\log_2 n\rceil-6)+(3\lceil\log_2 n\rceil+3)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$3\lceil\log_2 n\rceil+4$
Erlang	$(\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+3)\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+15\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$(\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+3)(4\lceil\log_2 n\rceil+2)+(6\lceil\log_2 n\rceil+9)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$3\lceil\log_2 n\rceil+5$
Maxwell-Boltzmann	$(\ln\left(\frac{1}{\epsilon}\right)+1)\left[(2n+8)\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+27\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$(\ln\left(\frac{1}{\epsilon}\right)+1)\left[(2n+8)(4\lceil\log_2 n\rceil-6)+(6\lceil\log_2 n\rceil+21)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]\cdot\frac{2}{\mathcal{F}}$	$3\lceil\log_2 n\rceil+6$

Family	Single-shot T-depth[3]	Single-shot T-count[4]	Success Probability
Normal	$\left((2n+8)\cdot\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+30\log_2(1/\epsilon_\theta)\right)\cdot\ln(1/\epsilon)$	$((8n+32)\lceil\log_2 n\rceil-8n+(9\lceil\log_2 n\rceil+21)\cdot\log_2(1/\epsilon_\theta)-12)\cdot\ln(1/\epsilon)$	$\frac{\mathcal{F}^2}{4}=7.37\%$
Exponential	$\left((n+3)\cdot\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+15\log_2(1/\epsilon_\theta)\right)\cdot\ln(1/\epsilon)$	$((4n+12)\lceil\log_2 n\rceil-2n+(6\lceil\log_2 n\rceil+9)\cdot\log_2(1/\epsilon_\theta)+2)\cdot\ln(1/\epsilon)$	$\frac{\mathcal{F}^2}{4}=0.583\%$
Rayleigh	$(2\ln\left(\frac{1}{\epsilon}\right)+1)\left[(n+4)\left(\left\lceil\log_2\frac{\log_2 n}{3}\right\rceil+2\right)+13.5\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$(2\ln\left(\frac{1}{\epsilon}\right)+1)\left[(n+5)(4\lceil\log_2 n\rceil-4)+3(\lceil\log_2 n\rceil+3.5)\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$\frac{\mathcal{F}^2}{4}=0.255\%$
Chi	$(2\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+4)\left(\left\lceil\log_2\frac{\log_2 n}{3}\right\rceil+2\right)+7.5\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$(2\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+4)(4\lceil\log_2 n\rceil-6)+(3\lceil\log_2 n\rceil+4.5)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$\frac{\mathcal{F}^2}{4}=3.00\%$
Chi-squared	$(\ln\left(\frac{1}{\epsilon}\right)+\frac{k}{2}-1)\left[(n+3)\left(\left\lceil\log_2\frac{\log_2 n}{3}\right\rceil+2\right)+6\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$(\ln\left(\frac{1}{\epsilon}\right)+\frac{k}{2}-1)\left[(n+3)(4\lceil\log_2 n\rceil-6)+(3\lceil\log_2 n\rceil+3)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$\frac{\mathcal{F}^2}{4}=2.29\%$
Erlang	$(\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+3)\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+15\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$(\ln\left(\frac{1}{\epsilon}\right)+k-1)\left[(n+3)(4\lceil\log_2 n\rceil+2)+(6\lceil\log_2 n\rceil+9)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$\frac{\mathcal{F}^2}{4}=0.130\%$
Maxwell-Boltzmann	$(\ln\left(\frac{1}{\epsilon}\right)+1)\left[(2n+8)\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+27\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$(\ln\left(\frac{1}{\epsilon}\right)+1)\left[(2n+8)(4\lceil\log_2 n\rceil-6)+(6\lceil\log_2 n\rceil+21)\cdot\log_2\left(\frac{1}{\epsilon_\theta}\right)\right]$	$\frac{\mathcal{F}^2}{4}=1.27\%$

[1, 2] The end-to-end T-depth/count utilizing amplitude amplification;
[3, 4] The single-shot T-depth/count without amplitude amplification that should be multiplied by $\frac{4}{\mathcal{F}^2}$.

TABLE III. Fault-tolerant quantum resource for state preparation of normal distribution

Method	Time-to-Solution	End-to-End T-Count	Ancillas
Re-parameterization[1]	$(20n+20)\lfloor\log_2(n-1)\rfloor+50n+3L\cdot\log_2(1/\epsilon_\theta)+20$	$\frac{21}{2}n^2+\frac{161}{2}n+3nL\cdot\log_2(1/\epsilon_\theta)-84\log_2 n-49$	n
Grover-Rudolph/Möttönen[2]	$3\log_2(1/\epsilon_\theta)\cdot(2^n-1)$	$3\log_2(1/\epsilon_\theta)\cdot(2^n-1)$	0
SI-PQC[3]	$\left((2n+8)\cdot\left(\left\lceil\log_2\frac{\log_2 n+2}{3}\right\rceil+2\right)+30\log_2(1/\epsilon_\theta)\right)\cdot\frac{2\ln(1/\epsilon)}{\mathcal{F}}$	$((8n+32)\lceil\log_2 n\rceil-8n+(9\lceil\log_2 n\rceil+21)\cdot\log_2(1/\epsilon_\theta)-12)\cdot\frac{2\ln(1/\epsilon)}{\mathcal{F}}$	$2\lceil\log_2 n\rceil+5$

[1] The re-parameterization method estimated when $L=n$ [27];
[2] The Grover-Rudolph method with Möttönen's uniformly controlled rotation synthesis [69];
[3] For a canonical interval $[\mu-3\sigma,\mu+3\sigma]$, $\mathcal{F}$ is 0.543.

reduction in dimension.

III. DISCUSSION

This paper proposes an efficient quantum state preparation and learning protocol named statistics-informed parameterized quantum circuit (SI-PQC). We introduce the maximum entropy principle to establish a general relationship between quantum state preparation and statistical symmetry in data. We then leverage this prior information to design SI-PQC. SI-PQC can improve the preparation and learning of statistical models, mitigating the potential input bottleneck and training problems for many practical quantum algorithms.

As an efficient state-preparation subroutine, SI-PQC avoids costly pre-processing procedures and enables in-depth circuit optimization. This is essential for quantum practicality, especially in real-time scenarios, e.g., initial state preparation in online machine learning and quantum-enhanced Monte Carlo integration for pricing financial derivatives or evaluating financial risk [24–28].

As a trainable and knowledge-informed quantum circuit, SI-PQC can improve the generalization ability, scalability, and trainability of QML algorithms, such as Gaussian mixture models and physics-informed neural networks [23, 33, 34]. Furthermore, these parameters are learned simultaneously to reveal the intrinsic statistical properties within the given model, promising enhanced theoretical interpretability.

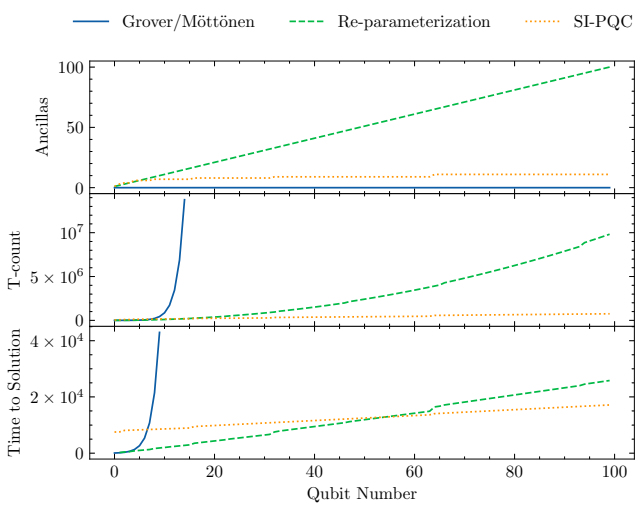

FIG. 8. **Quantum Resource Comparison for Preparing a Normal Distribution.** The blue solid line represents the Grover/Möttönen method, the green-colored dashed line represents the re-parameterization method, and the orange dotted line represents the SI-PQC method. The numerical results suggest that SI-PQC achieves an optimal quantum resource overhead for large problem size (n), and this asymptotic advantage can be diminished by a large prefactor for small n compared to the re-parameterization method.

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In principle, SI-PQC can be extended to load and simulate time series, develop quantum algorithms for non-parametric and semi-parametric statistical tests, and extract information from unknown intermediate quantum states of iterative quantum algorithms. Furthermore, there might be an interesting connection between SI-PQC and quantum state tomography/verification, wherein the maximum entropy principle has been introduced to reduce consumption of the quantum states [70]. We leave these interesting problems for future work.

IV. METHODS

Analytical Methods. We use block-encoding to embed a general linear operator A into a larger unitary matrix U_A as its upper left sub-matrix [71]. QSVT is then implemented to approximate the desired function via a sequence of subspace rotations [53]. We also use the linear combination of unitaries (LCU) technique [35] to implement the superposition of different unitaries. Therein, controlled unitary operators are implemented by conditioning on ancillary qubits. More technical details and proofs can be found in the Supplementary Information.

For the end-to-end quantum resource analysis, we consider qubit number and T-count. T-count is a standard characterization metric in the fault-tolerant era, as it is not transversal for many quantum error-correcting codes. More specifically, we consider end-to-end error analysis, combined with T+Clifford decomposition of the multi-controlled X gates and rotation gates, to tailor the resource requirements.

Numerical Methods. We demonstrate our quantum state preparation and learning protocol using standard circuit model quantum algorithms. All SI-PQC circuits were constructed on QPanda (v3.8.3) [72]. Qpanda is an open-source quantum computing software development framework provided by *Origin Quantum Computing* to build, run, and optimize quantum algorithms. We used a noise-free state vector simulator `pyqpanda.CPUQVM().getqstate()`. Because the qubit number and circuit depth are both large, we found that PyQpanda's statevector simulator is flexible and efficient. In the learning task, we used the limited-memory BFGS optimizer from SciPy.optimize [73]. We use Kullback–Leibler (KL) divergence to characterize the distance between the target state and the SI-PQC output state. KL divergence, also known as relative entropy, of two distributions is defined as:

$$D_{KL}(\mathbb{P},\mathbb{Q}) = \sum_x \mathbb{P}(x)\log\left(\frac{\mathbb{P}(x)}{\mathbb{Q}(x)}\right), \tag{28}$$

wherein $\mathbb{P}$ and $\mathbb{Q}$ are two distributions.

DATA AVAILABILITY

Data sharing is not applicable to this article as no datasets were generated or analysed during the current study.

ACKNOWLEDGEMENT

This work has been supported by the National Key Research and Development Program of China (Grant No. 2023YFB4502500) and the National Natural Science Foundation of China (Grant No. 12404564).

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FIGURE LEGENDS

Figure 1 **Schematic depiction of Statistics-Informed Parameterized Quantum Circuit (SI-PQC) via maximum entropy principle (MEP).** (a) MEP promises a unified formula to describe empirical data as linearly combined constraints from observables with maximized entropy. (b) SI-PQC achieves a universal expressiveness in the target space, and the output state stays therein during the optimization procedure (solid red arrow). While a problem-agnostic parameterized quantum circuit output state can often deviate from the target space when varying circuit parameters (dotted green arrow).

Figure 2 **Maximum entropy distribution loader.** (a) The quantum circuit for MEDL. (b-e) The numerical results to prepare Chi-squared (upper left), exponential (upper right), normal (lower left), and Rayleigh (lower right) distributions. The scatters (Exp.) represent the experimental results, and the lines (Theo.) are the theoretical discretization of the target distributions. The approximation error ϵ (the vertical distance between scatters and lines) originates from the imaginary time evolution and converges to zero exponentially quickly as the Taylor polynomial degree $d = \log 1/\epsilon$ grows. In this experiment, we are given the perfect prior knowledge about both the distribution law and the underlying parameters.

Figure 3 **State-preparation-pair for sparse linear combination of constraints.** Each (controlled) Y-rotation gate creates a superposition of amplitudes $\sqrt{\lambda_k \|f_k\|_\infty}$ and $\sum_{j\geq k+1} \sqrt{\lambda_j \|f_j\|_\infty}$, and each CX gate resets the former qubit to state $|0\rangle$ to create a superposition of one-hot states.

Figure 4 **Weighted distribution mixer.** (a) The quantum circuit for the Weighted Distribution Mixer. The state preparation

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pair (P_L, P_R) in MEDL is substituted by an analog-encoded space controlled by the digital-encoded latent space. (b) The numerical result to prepare a 4-mixture of normal distributions with means $\mu = (1, 5, 10, 15)$, variances $\sigma^2 = (30.25, 16, 36, 36)$, and weights $\vec{w}_1 = (0.1, 0.2, 0.3, 0.4)$, $\vec{w}_2 = (0.2, 0.3, 0.4, 0.1)$, $\vec{w}_3 = (0.3, 0.4, 0.1, 0.2)$, $\vec{w}_4 = (0.4, 0.1, 0.2, 0.3)$. The scatters (Exp.) are the experimental results, and the lines (Theo.) are the theoretical discretization of the target distributions. In this experiment, we are given only partial prior information about the distribution law. The model uncertainty is described by the latent space, wherein the mixture weight corresponds to the probability of different configurations of parameters.

Figure 5 **Comparison with related works for preparing a mixture of distributions.** (a) For general preparing method, an N_θ times repetition of circuits is needed for N_θ-components LCU. (b) For WDM method in this work, the circuit is implemented once with totally $\mathcal{O}(d_\theta + \log N_\theta) \log (1/\epsilon\mathcal{F})$ additional depth.

Figure 6 **Learning Gaussian Mixture Model based on SI-PQC.** In this experiment, we are given prior knowledge about the distribution family, and its parameters and corresponding weights are learned from the data via a variational method. (a) Quantum states in the learning procedure. (b) The estimated means are extracted from the angles in the training procedure. (c) The estimated variances are extracted from the angles in the training procedure. (d) The KL divergence is used to characterize the distance from the target state.

Figure 7 **Quantum Risk Analysis based on SI-PQC.** (a) Empirical data and estimated risk distribution, X is the underlying asset price, and $\mathbb{P}(X)$ is the empirical/estimated probability. (b) Monitored observables that can be efficiently derived by classical incremental computing. Herein, the observables $\mathbb{E}[X]$ and $\mathbb{E}[\ln X]$ vary slowly and can be regarded as conservative constraints, while $\mathbb{E}[X^2]$ changes violently. Henceforth, one can model this empirical stochastic variable by a Maxwell-Boltzmann distribution $\mathbb{P}(x) = \sqrt{\frac{2}{\pi}} \frac{x^2}{a^3} \exp\left\{\left(\frac{-x^2}{2a^2}\right)\right\}$. (c) Quantum states are prepared by SI-PQC using prior information from classical monitors. The model parameters can be directly estimated as $a = \sqrt{\frac{\pi}{8}}\mathbb{E}[X]$. (d) The solid, pointed, and dashed lines are the empirical Value at Risk (VaR)s given levels $\alpha = 0.95, 0.85, 0.05$, respectively. The diamond, star, and triangle markers are the quantum VaRs given levels $\alpha = 0.95, 0.85, 0.05$, respectively.

Figure 8 **Quantum Resource Comparison for Preparing a Normal Distribution.** The blue solid line represents the Grover/Möttönen method, the green-colored dashed line represents the re-parameterization method, and the orange dotted line represents the SI-PQC method. The numerical results suggest that SI-PQC achieves an optimal quantum resource overhead for large problem size (n), and this asymptotic advantage can be diminished by a large prefactor for small n compared to the re-parameterization method.

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